The documentation and process conversion measures necessary to comply with this revision shall be completed by 19 July 2001.

INCH POUND

MIL-PRF-19500/382C
AMENDMENT 2
19 April 2001
SUPERSEDING
AMENDMENT 1
25 September 2000

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, LOW-POWER
TYPE 2N2944A, 2N2945A, AND 2N2946A
JAN, JANTX, JANTXV, AND JANS

This amendment forms a part of MIL-PRF-19500/382C, dated 28 July 1999 and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* TITLE, delete and substitute: "SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, LOW-POWER TYPE 2N2944A, 2N2945A, 2N2945AM AND 2N2946A JAN, JANTX, JANTXV, AND JANS".

1.4, h_{FE} $(inv)_1$, delete "I_B = -200 µA dc", and substitute: "I_E = -200 µA dc".

* 1.4, delete "I_B = -1 µA dc" and substitute "r_{ec} $(on)_2$ I_B = -1 mA dc". (This reverses a change made in amendment 1.)

PAGE 2

3.3, h_{FE} (inv), delete and substitute: "Forward-current transfer ratio except that the collector and emitter shall be interchanged in the test circuit, i.e., I_E /I_B ."

PAGE 5

4.5.3, delete and substitute:

"4.5.3 Forward-current transfer ratio (inverted connection). Method of test shall be in accordance with method 3076 of MIL-STD-750, except that all references to the collector and the emitter of the transistor shall be interchanged in the test circuit. Then: h_{FE} (inv) = I_E / I_B."

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PAGE 8

* TABLE I, subgroup 2, forward-current transfer ratio, symbol h_{FE1}, delete and substitute:

"

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - continued.						
Forward-current transfer ratio	3076	V_{CE} = -0.5 V dc; I_C = -1.0 mA dc	h_{FE1}			
2N2944A				100		
2N2945A				70		
2N2945AM				70	200	
2N2946A				50		

"

TABLE I, subgroup 2, forward-current transfer ratio (inverted connection), conditions column, delete "I_B = -200 µA dc"; and substitute "I_E = -200 µA dc".

PAGE 9

TABLE I, subgroup 3, forward-current transfer ratio (inverted connection), conditions column, delete, "I_B = -200 µA dc" and substitute "I_E = -200 µA dc".

* TABLE I, subgroup 4, small signal emitter-collector on-state resistance, conditions column, delete, "I_B = -1 µA dc" and substitute "I_B = -1 mA".

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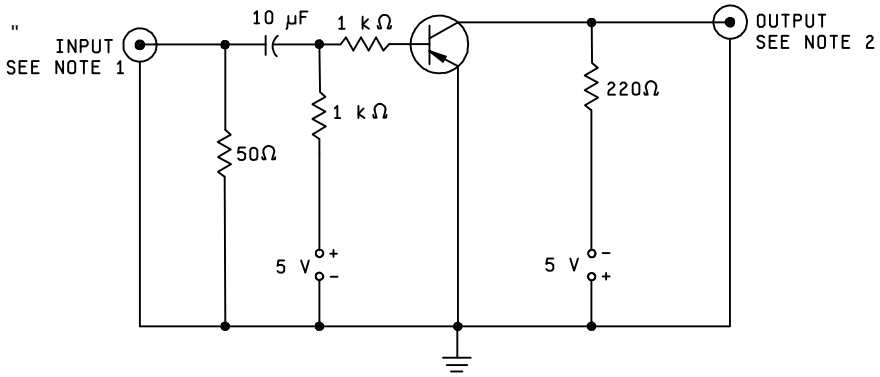
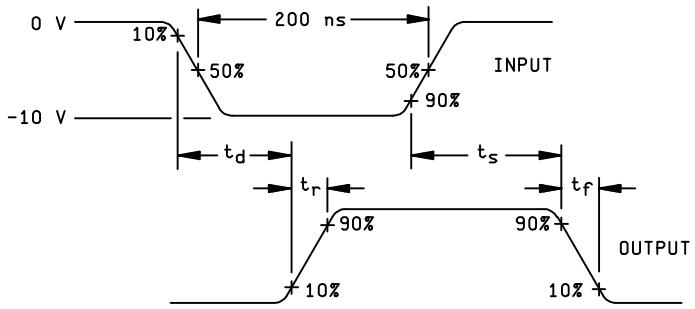
* FIGURE 4, delete and substitute:



TEST CIRCUIT



VOLTAGE WAVEFORMS

NOTES:
1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be ≤ 10 ns, duty cycle ≤ 2 percent. The input pulse width shall be 200 ns.
2. Output monitored with an oscilloscope with the following characteristics: Z_{in} ≤ 1 MΩ, t_r ≤ 1 ns.

FIGURE 4. Pulse response test circuit. "

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The margins of this amendment are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - CR
Navy - NW
Air Force - 11
NASA - NA
DLA – CC

Review activities:
Army - AV
Air Force - 19

Preparing activity:
DLA - CC

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